

End-to-end modeling of kilonova light curves and gravitational waveforms from neutron star mergers for possible constraints on nuclear physics parameters

by

Diego Fernando Hernández Porras

Submitted to the Department of Physics
in partial fulfillment of the requirements for the degree of

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ABSTRACT

Neutron stars are compact objects whose hydrostatic equilibrium is determined by the equation of state (EOS) of the dense matter in their interiors, which makes them natural laboratories for nuclear physics at densities inaccessible to terrestrial experiments. Although measurements of neutron-star masses, radii, and tidal deformabilities have constrained the pressure of dense matter, there is still no consensus on the composition of the stellar core, in particular on whether a phase transition to deconfined quark matter takes place. Binary neutron-star mergers (BNSMs) probe even higher densities and temperatures and, as demonstrated by the gravitational-wave event GW170817 and its associated kilonova, are a site of rapid neutron-capture (r-process) nucleosynthesis. In these events, the EOS shapes the hydrodynamic evolution of the merger and may therefore leave fingerprints both in the gravitational waveform and in the mass and initial thermodynamic conditions (e.g., density and entropy) of the ejected material. The subsequent nucleosynthesis in the ejecta, however, also depends on the properties of neutron-rich nuclei far from stability, including nuclear masses, β^- -decay and neutron-capture rates, and fission properties. Because dense-matter and nuclear-physics uncertainties affect the same observables, disentangling them is essential to determine what current and future multimessenger observations can reveal about dense matter. In this work, we perform a joint analysis of the gravitational-wave and electromagnetic signatures of BNSMs to investigate whether a hadron-quark phase transition leaves an imprint on the gravitational-wave signal and on the ejected material, whose mass and composition determine the radioactive heating that powers the kilonova. To this end, we perform general-relativistic hydrodynamic simulations of an equal-mass BNSM with the Einstein Toolkit, using a hadronic EOS, a hybrid EOS that coincides with it below the onset of a first-order hadron-quark phase transition, and an alternative hadronic EOS. Since the inspiral is identical for the hadronic and hybrid models by construction, post-merger differences can be attributed to the phase transition, while the alternative hadronic EOS allows us to assess whether the combined gravitational-wave and electromagnetic signatures can distinguish a phase transition from a change in the stiffness of hadronic matter. We compare the post-merger gravitational-wave signals and the dynamical ejecta, whose thermodynamic evolution we post-process with the nuclear reaction network WinNet to obtain r-process abundances and radioactive heating rates, from which we compute kilonova light curves.

Finally, we quantify the impact of the phase transition relative to the variation of the hadronic EOS, to uncertainties in nuclear-physics inputs such as fission-fragment distributions and β^- -decay rates, and to numerical resolution.

Thesis supervisor: Neelima Govind Kelkar

Title: Professor of Physics

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Este trabajo respresenta el final de una de las etapas mas bonitas de mi vida. Llevaré con mucho cariño el recuerdo de todos mis amigos a donde me lleve la motivación.

Contents

<i>List of Figures</i>	9
<i>List of Tables</i>	11
1 Introduction	13
1.1 Dense matter and neutron-star equation of state	13
1.2 Binary neutron-star mergers as multimessenger laboratories	13
1.3 Open problems: phase transitions and nuclear-physics uncertainties	13
1.4 Objectives and research questions	14
1.5 Structure of the thesis	14
2 Dense Matter and Neutron-Star Structure	15
2.1 Equation of state of cold, beta-equilibrated matter	15
2.2 Hadron–quark phase transitions and hybrid equations of state	15
2.3 Thermal effects in the equation of state	15
2.4 Stellar structure: TOV equations and tidal deformability	15
2.5 Observational constraints	15
3 Binary Neutron-Star Mergers	17
3.1 Merger phases and remnant fate	17
3.2 Gravitational waves: inspiral tidal effects and post-merger spectrum	17
3.3 Mass ejection: tidal, shock-driven, and disk-wind ejecta	17
3.4 Weak interactions and the electron fraction of the ejecta	17
3.5 Electromagnetic counterparts	17
4 Numerical Relativistic Hydrodynamics	19
4.1 3+1 decomposition and evolution formulation	19
4.2 General-relativistic hydrodynamics in conservative form	19
4.3 The Einstein Toolkit: initial data, mesh refinement, atmosphere	19
4.4 Gravitational-wave extraction	19
4.5 Ejecta diagnostics and unbound-matter criteria	19
5 r-Process Nucleosynthesis	21
5.1 Nuclear statistical equilibrium and freeze-out	21
5.2 The r-process: neutron-to-seed ratio, fission cycling	21
5.3 Reaction-network equations and numerical solution	21
5.4 Nuclear inputs: masses, decay rates, neutron capture, fission	21

5.5	Radioactive heating and thermalization	21
6	Kilonova Light Curves	23
6.1	Radiative transfer and the diffusion regime	23
6.2	Opacities of r-process matter	23
6.3	Light-curve modeling	23
7	Methods	25
7.1	Hadronic and hybrid equation-of-state models	25
7.2	Binary neutron-star simulation setup	25
7.3	From simulation to trajectories: ejecta extraction and expansion	25
7.4	Network setup and nuclear-physics variations	25
7.5	Kilonova model setup	25
7.6	Gravitational-wave analysis and detectability metrics	25
8	Results	27
8.1	Stellar properties of the equation-of-state models	27
8.2	Numerical accuracy and convergence	27
8.3	Merger dynamics and remnant evolution	27
8.4	Gravitational-wave signals	27
8.5	Dynamical-ejecta properties	27
8.6	Nucleosynthesis and heating rates	27
8.7	Kilonova light curves	27
8.8	Joint GW EM signatures	27
8.9	Uncertainty budget: EOS, nuclear physics, and resolution	27
8.10	Detectability with current and future facilities	27
9	Discussion	29
9.1	Comparison with previous studies	29
9.2	Limitations and model assumptions	29
10	Conclusions and Outlook	31
A	Units and conventions	33
B	Convergence tests	35
C	Network and equation-of-state configuration details	37

List of Figures

List of Tables

Chapter 1

Introduction

Here we will responds:

1. What question I am trying to answer?
Can the joint analysis of GW and EM signals encode relevant information about the existence of the phase transition inside a neutron star?
2. Why is this question open an relevant?
The possible confirmation or dismiss of quark-gluon plasma inside neutron stars could help us stablish the structure of neutron stars and test theoretical results in that regime.
3. What will I do to contribute to this question?
This thesis is just a prove of concept and a first approach to use the two different signals from a merger to try to identify possible encode information about the nature of the matter in the merger.

1.1 Dense matter and neutron-star equation of state

This section concentrates on the context of the second question. Here we will have a overview of the known equations of state, what they account for and their limitations.

1.2 Binary neutron-star mergers as multimessenger laboratories

This section is the state of the art of current efforts to use compact object mergers, in particular neutron star mergers, as test for theoretical develoment.

1.3 Open problems: phase transitions and nuclear-physics uncertainties

This section as continuation of the first one states why is so difficult to calculate the equation of state of matter inside a neutron star.

1.4 Objectives and research questions

This section focus on the first question, it states what question am i answering. Are combine GW and EM be usefull for testing theretical results?

1.5 Structure of the thesis

This is and overall review of what will i do in this thesis (answer to question 3)

Chapter 2

Dense Matter and Neutron-Star Structure

- 2.1 Equation of state of cold, β -equilibrated matter
- 2.2 Hadron–quark phase transitions and hybrid equations of state
- 2.3 Thermal effects in the equation of state
- 2.4 Stellar structure: TOV equations and tidal deformability
- 2.5 Observational constraints

Chapter 3

Binary Neutron-Star Mergers

- 3.1 Merger phases and remnant fate
- 3.2 Gravitational waves: inspiral tidal effects and post-merger spectrum
- 3.3 Mass ejection: tidal, shock-driven, and disk-wind ejecta
- 3.4 Weak interactions and the electron fraction of the ejecta
- 3.5 Electromagnetic counterparts

Chapter 4

Numerical Relativistic Hydrodynamics

- 4.1 3+1 decomposition and evolution formulation
- 4.2 General-relativistic hydrodynamics in conservative form
- 4.3 The Einstein Toolkit: initial data, mesh refinement, atmosphere
- 4.4 Gravitational-wave extraction
- 4.5 Ejecta diagnostics and unbound-matter criteria

Chapter 5

r-Process Nucleosynthesis

- 5.1 Nuclear statistical equilibrium and freeze-out
- 5.2 The r-process: neutron-to-seed ratio, fission cycling
- 5.3 Reaction-network equations and numerical solution
- 5.4 Nuclear inputs: masses, decay rates, neutron capture, fission
- 5.5 Radioactive heating and thermalization

Chapter 6

Kilonova Light Curves

6.1 Radiative transfer and the diffusion regime

6.2 Opacities of r-process matter

6.3 Light-curve modeling

Chapter 7

Methods

7.1 Hadronic and hybrid equation-of-state models

7.2 Binary neutron-star simulation setup

7.3 From simulation to trajectories: ejecta extraction and expansion

7.4 Network setup and nuclear-physics variations

7.5 Kilonova model setup

7.6 Gravitational-wave analysis and detectability metrics

physical mechanisms that produce the messengers. This is the main part of the chapter, and it is divided into three sections, each of which is further subdivided into subsections and subsubsections.

Chapter 8

Results

8.1 Stellar properties of the equation-of-state models

8.2 Numerical accuracy and convergence

8.3 Merger dynamics and remnant evolution

8.4 Gravitational-wave signals

8.5 Dynamical-ejecta properties

8.6 Nucleosynthesis and heating rates

8.7 Kilonova light curves

8.8 Joint GW EM signatures

8.9 Uncertainty budget: EOS, nuclear physics, and resolution

8.10 Detectability with current and future facilities

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Chapter 9

Discussion

9.1 Comparison with previous studies

9.2 Limitations and model assumptions

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Chapter 10

Conclusions and Outlook

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Appendix A

Units and conventions

Appendix B

Convergence tests

Appendix C

Network and equation-of-state configuration details